Subject: ENVIRONMENTAL IMPACT ASSESSMENT

Subject Code: 23A01504c

UNIT - IV

Environmental Audit

I. Introduction

An environmental audit is a systematic, documented, periodic and objective evaluation of how well an organisation, its management and its equipment are performing in protecting the environment.

Definition (ICC, 1989) "A management tool comprising a systematic, documented, periodic and objective evaluation of how well environmental organisation, management and equipment are performing, with the aim of helping to safeguard the environment by facilitating management control of environmental practices and assessing compliance with company policies, including regulatory requirements and standards."

- **In simple words** an environmental audit is a health check-up of a company's environmental performance.
- It is done on existing activities, while EIA is done before a project starts.
- Audit is a tool for management; it checks compliance, finds weaknesses and suggests improvements.
- In India, Environmental Statement (Form V) must be submitted every year by 30 September to the State Pollution Control Board, under the Environment (Protection) Rules, 1992 (introduced in 1993). This is a statutory form of environmental audit.

Audit versus EIA

Point	EIA	Environmental audit
Time	Before project	During running of project

Purpose	Predict impact	Verify actual performance and compliance
Basis	Predictions and models	Records, site inspection, measurements
Result	Clearance decision	Audit report and action plan

2. Environmental Audit and Environmental Legislation

- Laws set standards for emission, effluent, noise, waste and safety. Audit checks whether the industry follows them.
- **Main legislation covered** Water Act 1974, Air Act 1981, Environment Protection Act 1986, Hazardous Waste Rules, Noise Pollution Rules 2000, Biomedical Waste Rules, Solid Waste Rules and conditions of the Consent to Establish and Consent to Operate.
- **Audit also checks** EIA clearance conditions, EMP implementation, and ISO 14001 requirements.
- Failure to comply may lead to fines, closure notices, prosecution and loss of reputation.
- A regular audit helps the industry to find and correct violations before the authorities do.
- Government has also made audit mandatory in different forms, such as the Environmental Statement, hazardous waste returns and the Energy audit under the Energy Conservation Act 2001.
- Third party audit by recognised auditors is encouraged by the pollution control boards and in the Charter on Corporate Responsibility for Environmental Protection (CREP).

3. Objectives of Environmental Audit

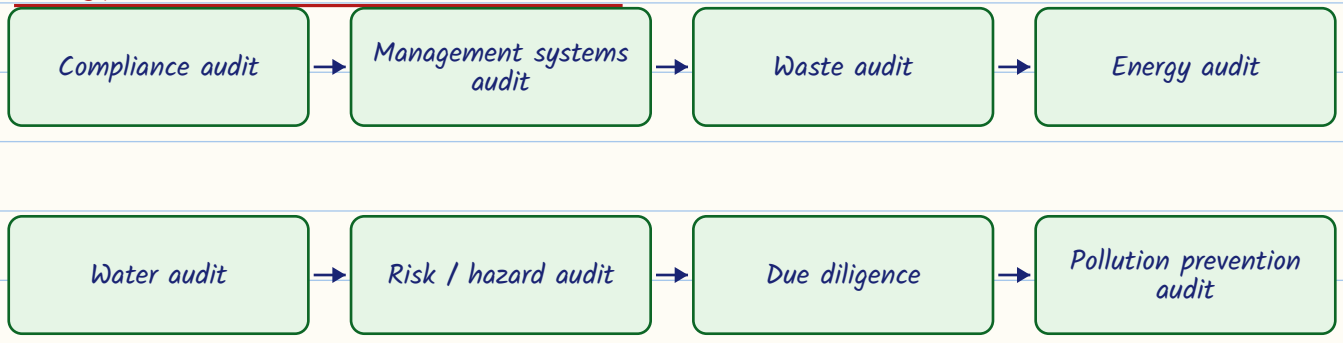
- To check compliance with laws, standards and permit conditions.
- To assess the effectiveness of pollution control equipment and the

environmental management system.

- To identify environmental risks and liabilities early.
- To find chances for saving raw materials, water, energy and money (waste minimisation and cleaner production).
- To improve environmental awareness and training among employees.
- To check that the company's own environmental policy is being followed.
- To provide reliable information to management, the public and regulators.
- To improve the public image of the organisation and relationship with the community.
- To help in insurance, loans and in buying or selling of properties (due diligence).
- To support continual improvement.

Benefits cost saving, legal safety, better image, fewer accidents, better resource use, better management of waste.

4. Types of Environmental Audit



- **Compliance audit** checks whether the plant follows all laws, rules, permits and standards. It is the most common type.
- **Environmental Management System (EMS) audit** checks whether the system (policy, planning, implementation, checking, review) as per ISO 14001 is working.

- **Waste (and emission) audit** studies the sources, types and quantities of waste, and ways for reduction, reuse and recycling.
- **Energy audit** studies the energy use and finds saving options.
- **Water audit** studies the water balance, losses, reuse and recycling.
- **Risk or hazard audit** studies the chances of accidents and the safety of storage, handling and emergency plans.
- **Due diligence (transactional) audit** done before purchase, merger or sale of a company or land to find hidden environmental liabilities.
- **Pollution prevention (minimisation) audit** finds ways to reduce pollution at the source.
- **Health and safety audit** checks occupational health and safety systems.
- **Product / life cycle audit** studies environmental effect of a product from raw material to disposal.

Based on who does it

- **Internal audit (first party)** done by the company's own trained staff.
- **External audit (second party)** done by customers or buyers.
- **Third party audit** done by independent outside auditors or certifying bodies. More credible.

Based on scope

- Site audit, corporate audit, issue specific audit, and follow up audit.

5. Audit Protocol

Audit protocol is a written set of procedures, checklists and questions, prepared beforehand, which guides the audit team on what to examine and how to record findings.

- It gives a uniform and consistent approach, and makes sure no important area is missed.

- It is used as a working document and its filled form becomes the evidence.

Contents of a protocol

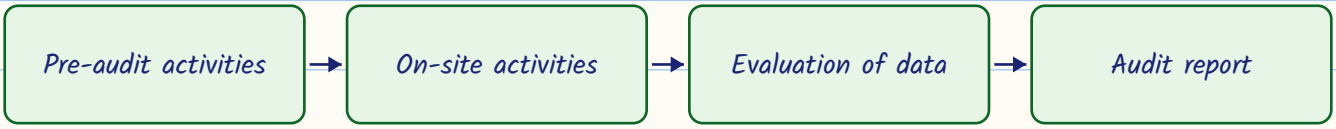
- Audit objective, scope and criteria (which laws and standards).
- **List of areas to be audited** raw material storage, process, utilities, air emissions, wastewater, solid and hazardous waste, noise, safety and emergency, records.
- Checklists and questionnaires for each area.
- Sampling and measurement methods.
- **Documents to be reviewed** consents, returns, monitoring reports, maintenance records, licences.
- Interview list of persons to meet.
- Format for recording observations and for the report.
- Rules of confidentiality and ethics.

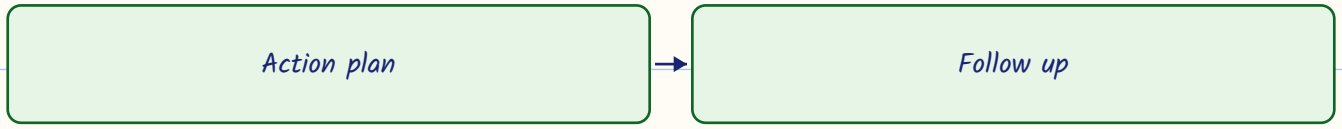
Sample questions in a checklist

- Is a valid Consent to Operate available? Are the limits followed?
- Is the effluent treatment plant working and are records kept daily?
- Are hazardous wastes labelled, stored and sent only to authorised recyclers?
- Are stack heights and monitoring facilities as per norms?
- Are employees trained for emergency?

6. Stages of Environmental Audit

The audit is carried out in three main stages: Pre-audit, Onsite audit (activities) and Post-audit.





6.1 Pre-audit (planning) activities

- Decide the objective, scope, depth and time of the audit.
- Select the audit team and the team leader; make sure they are independent and qualified.
- Inform the management and the plant personnel and fix the schedule.
- **Collect background information** site layout, process flow diagram, products, raw materials, past reports, licences, previous audit findings, complaints.
- Prepare the audit protocol, checklists and questionnaire.
- Plan the sampling, instruments and resources.
- Hold a preliminary meeting with the site managers.

6.2 Onsite activities

- Opening meeting with plant management to explain plan and procedure.
- **Understanding the management system and controls** review of the policy, responsibilities, training and procedures (this is checked with questionnaires and interviews).
- Assessing strengths and weaknesses of the internal controls.
- Physical inspection (site walk through) of the process, storage, pollution control units, drains, waste yard and housekeeping.
- **Collecting audit evidence** observation, interviews, document and record checks, sampling and measurements.
- **Verifying the evidence** cross checking and confirming findings with more than one source.
- Evaluating the findings against the criteria (laws, standards, company policy).
- Closing meeting, in which preliminary findings are discussed with management.

Evidence types physical (observation and samples), documentary (records and reports), testimonial (statements from persons) and analytical (laboratory results and calculations).

6.3 Post-audit activities

- Evaluation of audit data and preparation of the audit report.
- Management response and action plan with time schedule and responsibility.
- Follow up audit to check that the corrective actions have been carried out.

7. Evaluation of Audit Data

- All data from checklists, observations, measurements and records are collected and organised area wise.
- The findings are compared with the audit criteria – legal limits, permit conditions and company standards.
- **Each finding is classified** compliance, non-compliance (major or minor), or an observation / opportunity for improvement.
- Cause of each non-compliance is found, for example lack of training, equipment failure or poor maintenance.
- Findings are checked by different sources to be sure they are correct and fair.
- Risk ranking is done by how serious the effect is and how likely it is to occur, so that priority is given to the most severe items.
- Measured values (stack emissions, effluent quality, noise) are compared with standards in a table.
- Material balance and water balance are checked to find losses and wastage.
- Trends are studied over time (for example, last three years of monitoring data).
- Good practices are also noted and appreciated.

Finding	Criteria	Status	Priority

Effluent BOD 45 mg/L	Limit 30 mg/L	Non-compliance	High
Stack PM 120 mg/Nm3	Limit 150	Compliant	-
Hazardous waste not labelled	Rules	Minor non-compliance	Medium
Water meter missing	Good practice	Observation	Low

8. Preparation of the Audit Report

The audit report is the final document that presents the findings, conclusions and recommendations to the management.

Qualities of a good report

- Clear, accurate, brief, objective, balanced and complete.
- Supported by evidence; written in simple language.
- Constructive; points out good practices and not only the faults.
- Submitted on time and kept confidential.

Format of the report

- Title page and contents.
- Executive summary.
- **Introduction** objective, scope, criteria, dates, team and the methodology.
- Description of the facility and its activities.
- **Audit findings** area wise strengths, non-compliances and observations.
- Conclusions.
- Recommendations and corrective action plan, with priority, responsible person, time and cost.
- **Annexures** checklists, test results, photographs, list of persons met and documents examined.
- Signatures of the audit team.
- The draft report is discussed with the management for factual accuracy before the final report is issued.

Note the audit report is the property of the organisation. Findings should be used to improve, not to blame.

9. Environmental Statement (Form V) – Key Points

- Submitted by every industry requiring consent under the Water or Air Act, each year before 30th September for the financial year ended 31st March.
- **Contents** water and raw material consumption, pollutants discharged, hazardous waste generated, solid waste, steps taken for pollution control, investment, and proposals for further improvement.
- It is a self audit report to the State Pollution Control Board.

9A. Qualities and Duties of an Environmental Auditor

- Technical knowledge of the process, pollution control and environmental laws.
- Independence and honesty; no conflict of interest.
- Good observation, questioning and listening skills.
- Ability to write clear reports and communicate tactfully.
- Maintains confidentiality of company information.
- Uses a team approach: process engineer, chemist or analyst, safety expert and legal expert.
- **Duties** plan the audit, follow the protocol, collect and verify evidence, report truthfully and follow up.

9B. Specific Audits in Detail

Waste audit (step by step)

- **Step 1** draw the process flow diagram and mark every input and output.
- **Step 2** measure the quantity and quality of each waste stream – solid, liquid, gas and hazardous.

- **Step 3** prepare a material balance - Input = Product + Waste + Accumulation.
- **Step 4** find the sources of highest waste and the cost of disposal.
- **Step 5** suggest options - source reduction, reuse, recycling, recovery, treatment.
- **Step 6** estimate the cost and the payback period of each option and select the best.

Waste hierarchy Reduce at source -> Reuse -> Recycle -> Recover energy -> Treat -> Dispose safely (the last option).

Water audit

- Collect data on water sources, meters, uses in process, cooling, boiler, domestic and gardening.
- Prepare a water balance diagram: Total intake = Consumption + Losses + Wastewater discharged.
- Find leakages and overuse; check possibilities of recycle and reuse of treated water and rain water harvesting.
- **Example** A plant takes 1000 kL/day. Process uses 600, cooling loses 150 by evaporation, domestic 50 and wastewater discharged is 180. Total accounted = 980. Unaccounted loss = 20 kL/day (2 percent), probably leakage, which must be traced.

Energy audit

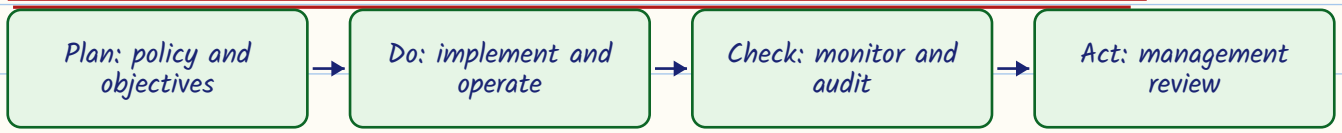
- Preliminary audit (walk through, using bills) and detailed audit (measurements with instruments).
- Study of power factor, motors, lighting, boilers, compressed air and heat recovery.
- Result is a list of energy saving measures with investment and payback.

Hazardous waste and safety audit

- Checks storage, labelling, manifest system, authorised transporters and disposal sites.

- Checks fire fighting, alarms, personal protective equipment, safety training, mock drills and the on-site emergency plan.

9C. ISO 14001 Based EMS Audit (Plan – Do – Check – Act)



- The auditor checks that the policy is approved by top management and known to all workers.
- Checks that significant environmental aspects and legal requirements are identified.
- Checks training, communication, document control and emergency preparedness.
- Checks monitoring records, calibration of instruments, non-conformity handling and internal audits.
- Management review must be done at planned intervals to give continual improvement.

9D. Advantages and Limitations of Environmental Audit

Advantages

- Early warning of problems and compliance gaps.
- Saves money by reducing waste, energy and water use.
- Improves safety and reduces accidents and liability.
- Builds confidence among regulators, banks, insurers and customers.
- Encourages employee participation and responsibility.
- Gives a base for ISO 14001 certification and for green marketing.

Limitations

- Needs time, money and trained persons.
- If done carelessly it becomes only a paper exercise.
- Fear of exposing violations may cause managers to hide information.
- Results depend on the quality of records and honesty of staff.
- Without follow up, the recommendations may not be implemented.

9E. Green Audit and Other Related Terms

- **Green audit** environmental audit of institutions like colleges, hospitals and offices - covers energy, water, waste, greenery and transport.
- **Social audit** checks the social effects of a project on workers and community.
- **Environmental performance evaluation (EPE)** uses indicators such as emission per tonne of product, water per unit and waste generated.
- **Life Cycle Assessment (LCA)** studies the environmental load of a product from cradle to grave.
- **Eco-labelling (Ecomark)** label for environment friendly products in India.

10. Key Points to Remember

- Environmental audit = systematic, documented, periodic and objective evaluation of environmental performance.
- **Types** compliance, EMS, waste, energy, water, risk, due diligence, pollution prevention; internal, external, third party.
- Protocol = checklists and procedures that guide the audit.
- Stages = pre-audit, onsite audit, post audit.
- **Onsite steps** understand system -> assess controls -> gather evidence -> evaluate -> report.
- Audit follow up ensures that recommendations are really implemented.

Important questions (1) Define environmental audit and explain its objectives. (2) Types of environmental audit. (3) What is an audit protocol? (4) Explain the stages of an environmental audit. (5) How is audit data evaluated and the report prepared?